

Anirudh Som

Contact Info:

Email: contact.anirudh.som@gmail.com **Website** [Website](#) **LinkedIn** [LinkedIn](#) **Google Scholar** [Google Scholar](#)

Work Authorization: U.S. Permanent Resident

Skills:

Programming – Python, Matlab, XML, Latex

Libraries & Frameworks – PyTorch, TensorFlow, Transformers, Scikit-learn, Pandas, OpenCV, DeepSpeed

LLMs & VLMs – GPT, LLaMa, Mixtral, Mistral, Open LLaMa | RAG, RLHF, SFT, LoRA, Perceiver, PEFT

Infrastructure – Docker, Kubernetes, Gitlab, Artifactory, AWS, GPU cluster training, FSDP, ZeRO

Data Handling – Document parsing, OCR pipelines, Structured extraction from unstructured sources

Education:

Ph.D., Electrical Engineering | Computer Vision & Machine Learning – Arizona State University, Tempe, AZ

Professional Experience:

- **Vice President - Applied AI/ML** – JPMorganChase, New York, NY **2025 – Present**
 - Part of the Payments - Trust & Safety team in the Commercial & Investment Banking unit.
 - Large Payments Model (LPM): Built JPMC's first transformer-based foundation model for fraud detection in payment transactions.
 - Automated Clearing House (ACH): Improved precision and recall-by-volume by 27%, and recall-by-value by 64%, over production Catboost baseline.
 - Wire: Improved precision and recall-by-volume by 6% over Catboost baseline.
 - Serving: Co-built the GPU-optimized serving platform powering LPM, achieving sub-100ms inference at production scale (~40ms P90, ~60ms P99).
 - Optimizations: Optimized model architecture and training platform, reducing both training and inference time by 50-75% and model size by ~70%, with little to no drop in model performance.
- **Advanced Computer Scientist** – SRI, Princeton, NJ **2020 – 2025**
 - CV & NLP Systems: Led development and deployment of transformer-based CV and LLM pipelines for real-time and offline video understanding, including scene segmentation, summarization, metadata extraction, and emotion-aware HMLs.
 - Document Intelligence: Built OCR+NLP pipelines and LLM systems to extract structured insights from unstructured medical and educational documents.
 - Mentorship & Leadership: Served as technical lead on federal and commercial AI projects, mentoring and supervising 2–4 junior ML engineers and interns.
 - Commercial Use Cases: Delivered AI prototypes and proof-of-concepts for Fortune 500 clients, including L'Oréal and Honda.
 - Product Collaboration: Collaborated with cross-functional teams to deploy and maintain ML solutions aligned with product goals and operational needs.

Key Projects:

- ARPA-H PARADIGM: CV+LLM-driven task guidance system for rural healthcare support.
- DARPA CCU: CV+LLM pipelines for cross-cultural awareness in dialogue systems.
- DARPA EDGE: Real-time emotion-aware HMI systems.
- NSF: Automated classroom collaboration assessment using multimodal architectures.

Pre-Doctoral Experience:

- **Digital Pathology Intern** – Roche Diagnostics, Santa Clara, CA **2019**

Image segmentation, federated learning in digital pathology.
- **Computation Intern** – Lawrence Livermore National Laboratory, Livermore, CA **2018**

Time-series deep learning for heart signal modeling.
- **Medical Imaging Intern** – Mayo Clinic, Phoenix, AZ **2015**

ML for cancer detection in CT/MRI.

Selected

Publications:

- [1] A. Som et al. "[MINDS: A Cross-cultural Dialogue Corpus for Social Norm Classification and Adherence Detection](#)". Accepted at the International Joint Conference on NLP & Asia-Pacific Chapter of the Association for Computational Linguistics (IJCNLP-AAACL) Findings, 2025.
- [2] A. Som et al. "[Demonstrations Are All You Need: Advancing Offensive Content Paraphrasing using In-Context Learning](#)". Accepted at the Association for Computational Linguistics (ACL) Findings, 2024.
- [3] A. Som, et al. "[Automated Student Group Collaboration Assessment and Recommendation System Using Individual Role and Behavioral Cues](#)". Accepted at the Frontiers in Computer Science Journal, 2021.
- [4] A. Som, et al. "[Perturbation Robust Representations of Topological Persistence Diagrams](#)". Accepted at the European Conference on Computer Vision (ECCV), 2018.

Complete List: [Google Scholar](#)

Patents:

- [1] [Machine Learning Model Prompt Demonstration Selection](#).
- [2] [Automated Identification of Necrotic Regions in Digital Images of Multiplex Immunofluorescence-Stained Tissue](#).
- [3] [Federated Learning System for Training Machine Learning Algorithms Without Accessing Patient Data to Protect Patient Privacy](#).
- [4] [Automated Collaboration Skills Assessment](#).